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Review of the recreational daily kina limit in fishery management area 1 (the east coast of the upper North Island) AND enabling the removal of sea urchins for the management or prevention of urchin barrens

Introduction

1. This submission addresses Fisheries New Zealand's (**FNZ**) proposed increase in recreational daily take limits for kina.
2. This submission also addresses FNZ's proposed introduction of a new special permit purpose under section 97 of the Fisheries Act 1996, "to allow persons or organisations to harvest, cull, or translocate sea urchins for the purpose of habitat restoration and/or prevention of urchin barrens."
3. Together, these proposed measures aim to reduce the extent and adverse effects of urchin barrens (including kina) and to facilitate ecosystem restoration, including kelp forest restoration.
4. The Environmental Defence Society (**EDS**) is an independent not-for-profit organisation conducting interdisciplinary policy research and litigation. It was established in 1971 with the purpose of improving environmental outcomes in Aotearoa New Zealand.

5. EDS has a special interest in the marine environment and recently completed the first phase of a multi-year project looking at options for future reform of the oceans management system.¹ This included, among other things, fisheries management. In 2018, EDS led an in-depth review of the national fisheries management system and published findings in a report entitled *Voices from the Sea: Managing New Zealand's Fisheries*.² It has also sought to improve fisheries decision-making by submitting on proposals to set sustainability measures for the management of various wild fish stocks.³

Summary of submission

6. Sea urchin barrens pose a significant threat to marine ecology in Aotearoa New Zealand, arising from consistent and prolonged overgrazing of benthic vegetation, leading to reductions in marine habitat complexity, productivity, and biodiversity.⁴
7. EDS supports FNZ in its efforts to restore ecology and biodiversity in New Zealand's coastal waters, by reducing the extent and adverse effects of urchin barrens. In this regard, EDS is not, in principle, opposed to an increase in recreational daily kina take limits, nor the provision for harvest, culling, or translocation of urchin in non-commercial, recreational, or cultural capacities, provided appropriate conditions are implemented.
8. However, on their own, these measures fail to address the root causes of urchin barrens, being primarily the overfishing and removal of key sea urchin predators including snapper and crayfish. If FNZ is serious about restoring barren areas to durable biodiverse marine ecosystems, it must develop and implement management measures to address these root causes.

Proposed increase in recreational take limits

9. FNZ proposes an increase in recreational daily take limits for kina on the east coast of Te Ika-a-Māui / North Island as "one of many tools" in a broader management effort to reduce the extent and impact of kina barrens. FNZ "envision[s]" this effort to include "initiatives like rock lobster and snapper management" and "increased research efforts".⁵
10. EDS agrees that appropriate kina management must be integrated, and multi-pronged, and that one of these prongs could be an increase in recreational daily take limits for kina. However, EDS is concerned about the lack of evidence or detail of other management "initiatives" FNZ intends to implement. Specifically, without initiatives addressing overfishing of kina predators, an increase in recreational kina take limits is highly unlikely to achieve significant ecological improvements.
11. EDS suggests that such measures need to be supported by spatial protections, including fisheries closures and marine protected areas (**MPA**), around reefs susceptible to urchin barrens so that kina predators like large crayfish and snapper can re-establish. Studies conducted in north-eastern Aotearoa New Zealand indicate that while urchin barrens are

¹ Greg Severinsen and others, 2022, *The Breaking Wave: Oceans Reform in Aotearoa New Zealand*, Environmental Defence Society, Auckland, available from www.eds.org.nz

² Raewyn Peart, 2018, *Voices from the Sea: Managing New Zealand's Fisheries*, Environmental Defence Society, Auckland, available from www.eds.org.nz

³ Copies of EDS's recent submissions on a range of wild fish stocks are available from www.eds.org.nz

⁴ Kelsey Miller and Nick Shears, 2022, *The efficiency and effectiveness of different sea urchin removal methods for kelp forest restoration*, Restoration Ecology 31(1) <https://doi.org/10.1111/rec.13754>

⁵ Fisheries NZ Discussion Paper No: 2024/12, at paragraphs 2 and 112.

prevalent outside MPAs, nearby fully protected MPAs show lasting ecosystem recovery, pointing to overfishing as the primary cause of urchin barrens.⁶

12. EDS also shares mana whenua concerns about the potential for localised depletion of healthy kina populations.⁷ Collection from kina barrens is disincentivised, because kina located there are typically small and skinny, making them less appetising. This means that increasing daily bag limits may focus more effort on healthy reef systems and do little to address barrens.⁸
13. With this in mind, EDS prefers an increase in recreational daily take limits for kina from 50 to 100 (i.e. proposed 'Option 2'), provided spatial restrictions are implemented, and provided other effective measures like fisheries closures and MPAs are adopted to address overfishing of kina predators.

Sea urchin harvest, culling, and relocation

14. FNZ proposes the introduction of a new special permit purpose under s 97 of the Fisheries Act 1996, "to allow persons or organisations to harvest, cull, or translocate sea urchins for the purpose of habitat restoration and/or prevention of urchin barrens." This is intended to enable permits to be granted for removals of sea urchins from urchin barrens in non-commercial, recreational, or cultural capacities.
15. EDS notes that similar sea urchin management approaches adopted overseas (e.g. California and the Pacific Northwest) have been effective in restoring kelp forests and biodiversity. In Aotearoa New Zealand, large-scale sea urchin removals have also promoted rapid kelp recovery.⁹ However, as discussed above, the durability of restored areas largely depends on the implementation of measures to address overfishing of sea urchin predators, to control sea urchin reinvasion.¹⁰
16. The effectiveness of sea urchin removals also largely depends on appropriate conditions being set. EDS strongly agrees that removal permits should only be issued for specific areas where urchin barrens are present or where there is a serious risk of an urchin barren.
17. EDS is particularly concerned about the relocation and introduction of sea urchins from urchin barrens to healthy reef ecosystems. Rigorous analysis must be undertaken to ensure any harvest and/or predation of introduced sea urchins is adequate to maintain balance within the receiving environment.

Conclusion

18. EDS supports an increase in recreational daily take limits for kina from 50 to 100 per person per day. EDS also supports the enablement of sea urchin removals, for the purpose of habitat restoration and/or prevention of urchin barrens.

⁶ Ohad Peleg and others, 2023, *Long-term marine protection enhances kelp forest ecosystem stability*, Ecological Applications 33(7) <https://doi.org/10.1002/eap.2895>

⁷ Fisheries NZ Discussion Paper No: 2024/12, at paragraph 122.

⁸ Fisheries New Zealand Discussion Paper No: 2024/11 at paragraph 67

⁹ Kelsey Miller and others, 2023, *Large-scale one-off sea urchin removal promotes rapid kelp recovery in urchin barrens*, Restoration Ecology 32(1) <https://doi.org/10.1111/rec.14060>

¹⁰ Kelsey Miller and others, 2022, *Sea Urchin Removal as a Tool for Macroalgal Restoration: A Review on Removing "the Spiny Enemies"*, Frontiers in Marine Science 9 <https://doi.org/10.3389/fmars.2022.831001>.

19. EDS considers that such measures will not be effective in achieving a long term reduction in the extent of urchin barrens without other measures including spatial protections to enable predators to re-establish and the food web to rebalance as part of healthy ecosystems.
20. Critically, the restoration and durability of habitat health, complexity, productivity and biodiversity in the marine environment depends on the implementation of integrated measures like fisheries closures and MPAs to address overfishing of sea urchin predators.